

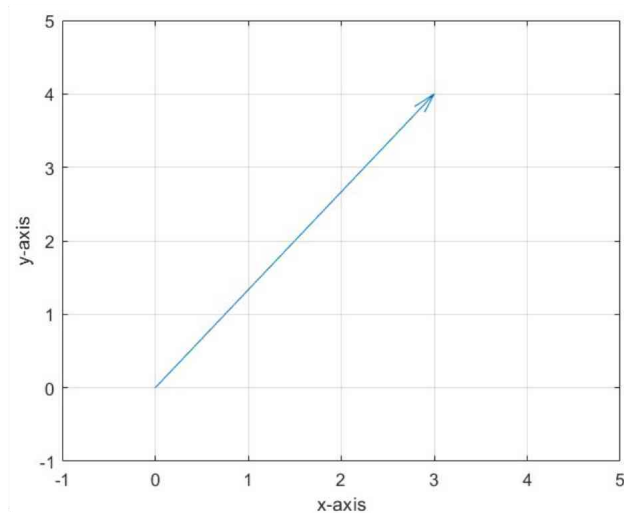
Practice Problem 1

1. A car is traveling 30 mph northeast.
 - What is the magnitude of the car's velocity?
 - 30mph
 - What is the direction of the car's velocity?
 - Northeast

Practice Problem 2

2. Consider a vector defined as $\bar{F} = 3\hat{a}_x + 4\hat{a}_y$

- Sketch the vector



- What is the magnitude of the vector?

- $|\bar{F}| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

Practice Problem 3

3. Given the following, solve the equations:

$$\bar{A} = 1\hat{a}_x + 2\hat{a}_y + 3\hat{a}_z, \quad \bar{B} = 4\hat{a}_x - 5\hat{a}_y + 4\hat{a}_z$$

- $\bar{C} = \bar{A} + \bar{B}$
- $\bar{D} = \bar{A} + 2\bar{B}$
- $\bar{E} = 3\bar{A} + \bar{B}$

$$\begin{aligned} C_x &= A_x + B_x \\ C_y &= A_y + B_y \\ C_z &= A_z + B_z \end{aligned}$$

Practice Problem 3: Solution

- $\bar{C} = \bar{A} + \bar{B}$
 - $C_x = A_x + B_x = 1 + 4 = 5$
 - $C_y = A_y + B_y = 2 - 5 = -3$
 - $C_z = A_z + B_z = 3 + 4 = 7$
 - $\bar{C} = 5\hat{a}_x - 3\hat{a}_y + 7\hat{a}_z$
- $\bar{D} = \bar{A} + 2\bar{B}$
 - $D_x = A_x + 2B_x = 1 + 8 = 9$
 - $D_y = A_y + 2B_y = 2 - 10 = -8$
 - $D_z = A_z + 2B_z = 3 + 8 = 12$
 - $\bar{D} = 9\hat{a}_x - 8\hat{a}_y + 12\hat{a}_z$

Practice Problem 3: Solution

- $\bar{E} = 3\bar{A} + \bar{B}$
 - $E_x = 3A_x + B_x = 3 + 4 = 7$
 - $E_y = 3A_y + B_y = 6 - 5 = 1$
 - $E_z = 3A_z + B_z = 9 + 4 = 13$
 - $\bar{E} = 7\hat{a}_x + 1\hat{a}_y + 13\hat{a}_z$

Practice Problem 4

4. Given the following, solve the equations:

$$\bar{A} = 1\hat{a}_x + 4\hat{a}_y + 4\hat{a}_z, \quad \bar{B} = 2\hat{a}_x - 3\hat{a}_y - 1\hat{a}_z$$

- $\bar{C} = \bar{A} - \bar{B}$

- $\bar{D} = 4\bar{A} - 2\bar{B}$

- $\bar{E} = 2\bar{A} - \bar{B}$

$$C_x = A_x - B_x$$

$$C_y = A_y - B_y$$

$$C_z = A_z - B_z$$

Practice Problem 4: Solution

- $\bar{C} = \bar{A} - \bar{B}$
 - $C_x = A_x - B_x = 1 - 2 = -1$
 - $C_y = A_y - B_y = 4 + 3 = 7$
 - $C_z = A_z - B_z = 4 + 1 = 5$
 - $\bar{C} = -1\hat{a}_x + 7\hat{a}_y + 5\hat{a}_z$
- $\bar{D} = 4\bar{A} - 2\bar{B}$
 - $D_x = 4A_x - 2B_x = 4 - 4 = 0$
 - $D_y = 4A_y - 2B_y = 16 + 6 = 22$
 - $D_z = 4A_z - 2B_z = 16 + 2 = 18$
 - $\bar{D} = 0\hat{a}_x + 22\hat{a}_y + 18\hat{a}_z$

Practice Problem 4: Solution

- $\bar{E} = 2\bar{A} - \bar{B}$
 - $E_x = 2A_x - B_x = 2 - 2 = 0$
 - $E_y = 2A_y - B_y = 8 + 3 = 11$
 - $E_z = 2A_z - B_z = 8 + 1 = 9$
 - $\bar{E} = 0\hat{a}_x + 11\hat{a}_y + 9\hat{a}_z$

Practice Problem 5

5. Given the following, solve the equations:

$$\bar{A} = 3\hat{a}_x + 4\hat{a}_y + 1\hat{a}_z, \quad \bar{B} = 2\hat{a}_x + 3\hat{a}_y - 4\hat{a}_z$$

- $\bar{C} = \bar{A} \cdot \bar{B}$

$$\bar{A} \cdot \bar{B} = A_x B_x + A_y B_y + A_z B_z$$

- $\bar{D} = 2\bar{A} \cdot -\bar{B}$

Practice Problem 5: Solution

- $\bar{C} = \bar{A} \cdot \bar{B}$
 - $\bar{C} = (3)(2) + (4)(3) + (1)(-4)$
 - $\bar{C} = 6 + 12 - 4$
 - $\bar{C} = 14$
- $\bar{D} = 2\bar{A} \cdot -\bar{B}$
 - $\bar{D} = (6)(-2) + (8)(-3) + (2)(4)$
 - $\bar{D} = -12 - 24 + 8$
 - $\bar{D} = -28$